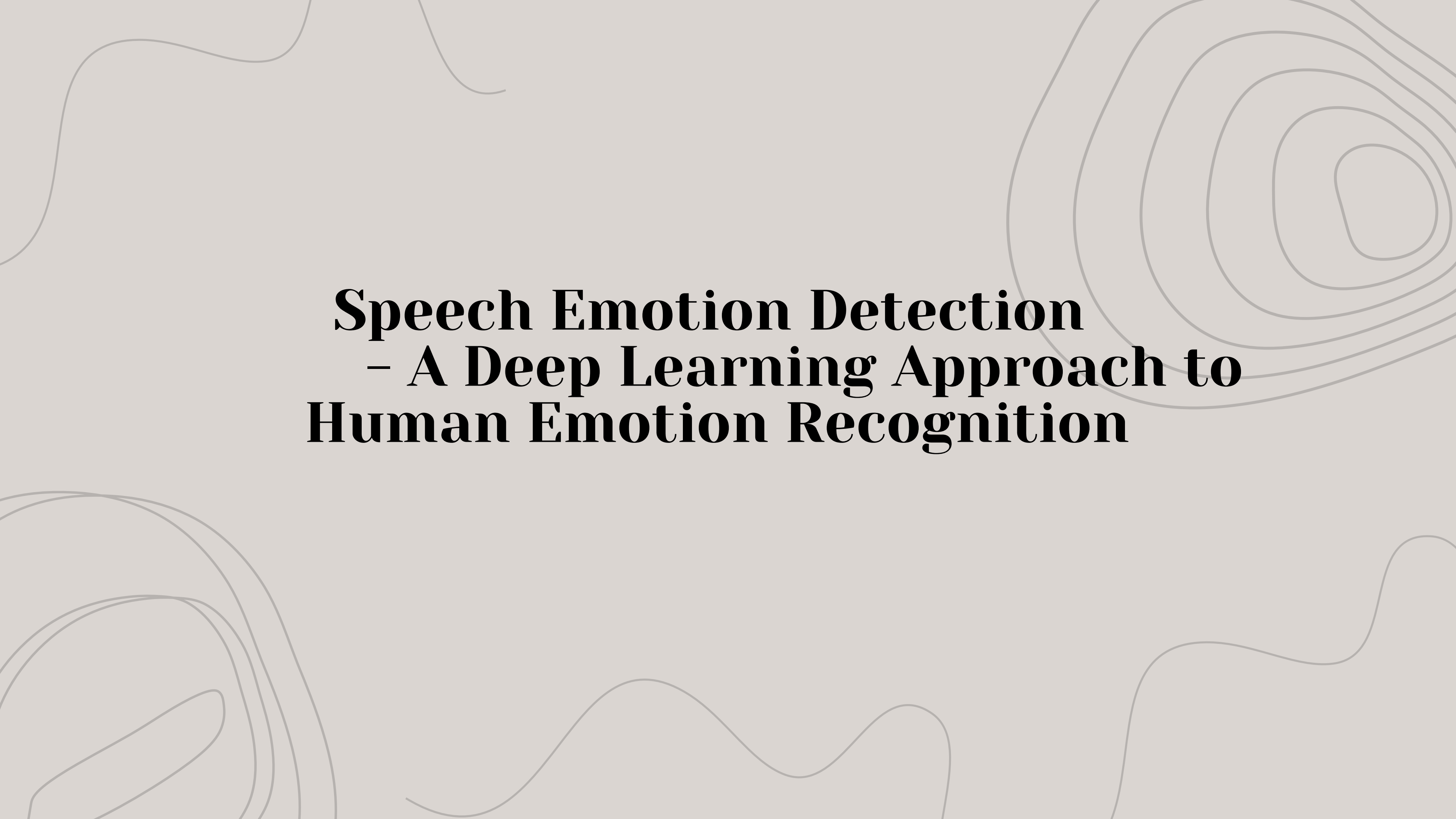




Speech Emotion Detection **– A Deep Learning Approach to** **Human Emotion Recognition**

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Abstract

This report details a deep learning approach for automatic speech emotion detection (SED), a key element in affective computing and human-computer interaction. The project develops a robust neural network for identifying six emotion categories: neutral, happy, sad, angry, fear, and disgust from audio speech signals with high accuracy across diverse conditions.

A hybrid CNN-BiLSTM-Transformer model is proposed: CNNs capture frequency-domain features from mel-spectrograms, while LSTMs model temporal dependencies. The complete pipeline includes audio preprocessing, log-mel spectrograms with delta and delta-delta coefficients (80 mel bins), and classification. Training utilizes augmentation techniques, cross-validation, and hyperparameter tuning. Evaluation is performed on benchmark datasets (RAVDESS, CREMA-D) using standard metrics, with adaptive, SNR-aware preprocessing to reduce the impact of noisy recordings.

The expected outcome is state-of-the-art recognition accuracy with real-time computational efficiency for practical deployment. This work demonstrates deep learning effectiveness for emotionally intelligent systems in areas such as mental health monitoring, customer service, adaptive learning, and human-robot interaction, contributing to more empathetic AI.

Aim

- Speech emotion detection plays a vital role in improving human-computer interaction, mental health analysis, and customer experience. However, current models achieve only about 70–80% accuracy due to noise, accent, and demographic variations. Hence, our goal is to develop a robust deep learning-based system that ensures accurate, generalizable emotion recognition across diverse real-world conditions.
- Develop a robust speech emotion detection system using advanced deep learning techniques for automatic classification of human emotions from audio speech signals.
- Achieve high accuracy and reliability across diverse speakers, recording conditions, and real-world scenarios.
- Create cross-demographic generalization that works effectively across different speakers, ages, genders, accents, and cultural backgrounds

Objectives

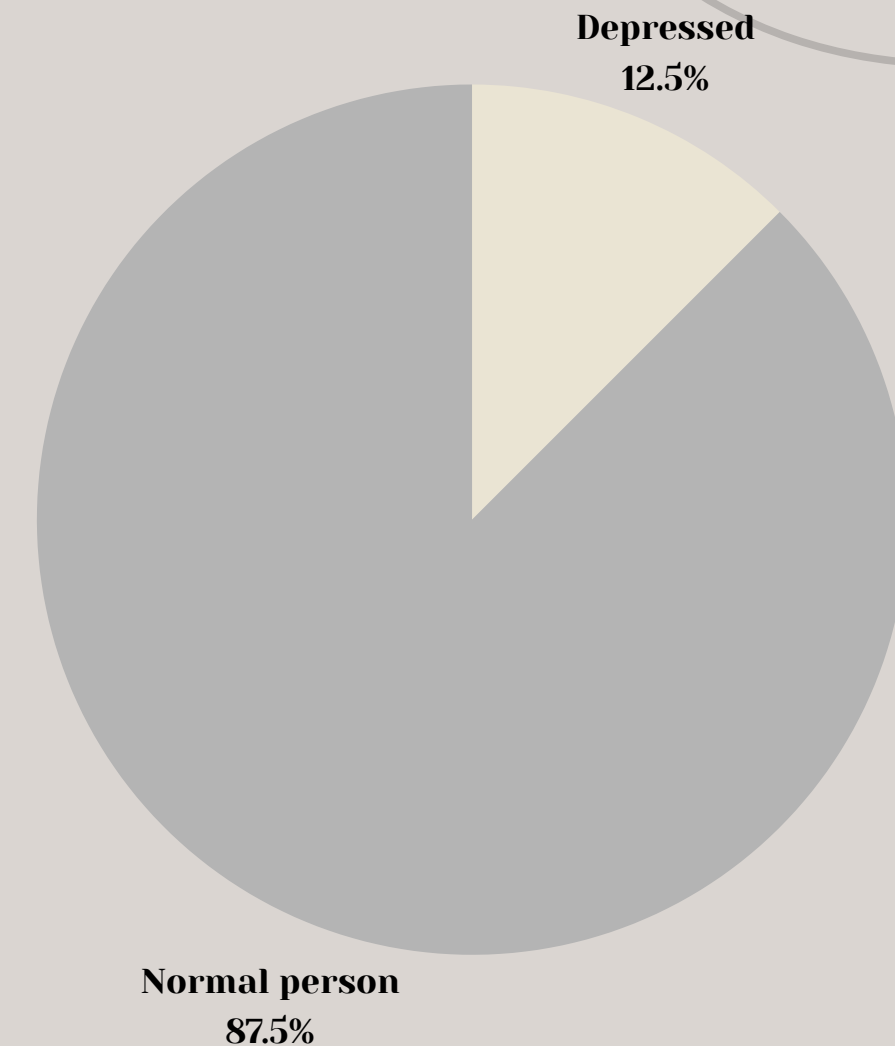
The core objectives include Advanced Architecture Implementation—specifically, deploying and evaluating implementing a hybrid CNN-BiLSTM-Transformer architecture that combines convolutional, recurrent, and attention-based components in a single model alongside Comprehensive Feature Engineering, which mandates the integration of MFCCs, spectral features, prosodic features, and mel-spectrograms to create rich input data. A critical goal is the use of Enhanced Training Methodologies, encompassing data augmentation, regularization, all geared toward the overarching aim of Performance Optimization: achieving state-of-the-art accuracy while guaranteeing real-time computational efficiency. Finally, the project aims for tangible deployment capabilities through both User-friendly Interface Development (making the system accessible to all users) and ensuring robust System Integration Capabilities via APIs and protocols for seamless deployment across various platforms.

Scope of Work

- This advanced system architecture leverages a Hybrid CNN-LSTM structure enhanced with Transformer-based attention mechanisms for robust sequence and feature learning. Its core capability lies in multi-scale feature integration coupled with advanced preprocessing, relying on key technologies like LibROSA, and the Transformers library for implementation. For Deployment, the system is designed with privacy-preserving local processing to support efficient execution. It prioritizes user experience through user-friendly interfaces and ensures data security via privacy-preserving local processing. The versatile Applications span across several domains, including Healthcare (mental health monitoring, elderly care), Education (adaptive learning, engagement tracking), Business (customer service automation, driver safety monitoring), and HCI (accessible interfaces, emotion-aware AI systems).

Motivation of Work

We are motivated by the challenge of non-intrusive mental health monitoring, a clear societal need. Globally, an estimated 5.7% of adults suffer from depression, and nearly 1 in 7 people live with a mental disorder. We have seen how difficult it is to detect the onset of distress early. Our goal is to deploy this technology to passively monitor a patient's voice during routine telehealth or elder care interactions, automatically flagging subtle changes in sadness or anxiety levels for clinicians, thus providing a critical, early-warning system and helping to address the massive burden of mental health conditions.



Literature Survey

Global-Local-Feature-Fused Driver Speech Emotion Detection for Intelligent Cockpit in Automated Driving

Author(s): Wenbo Li, Jiyong Xue, Ruichen Tan, Cong Wang, Zejian Deng, Shen Li, Gang Guo, Dongpu Cao

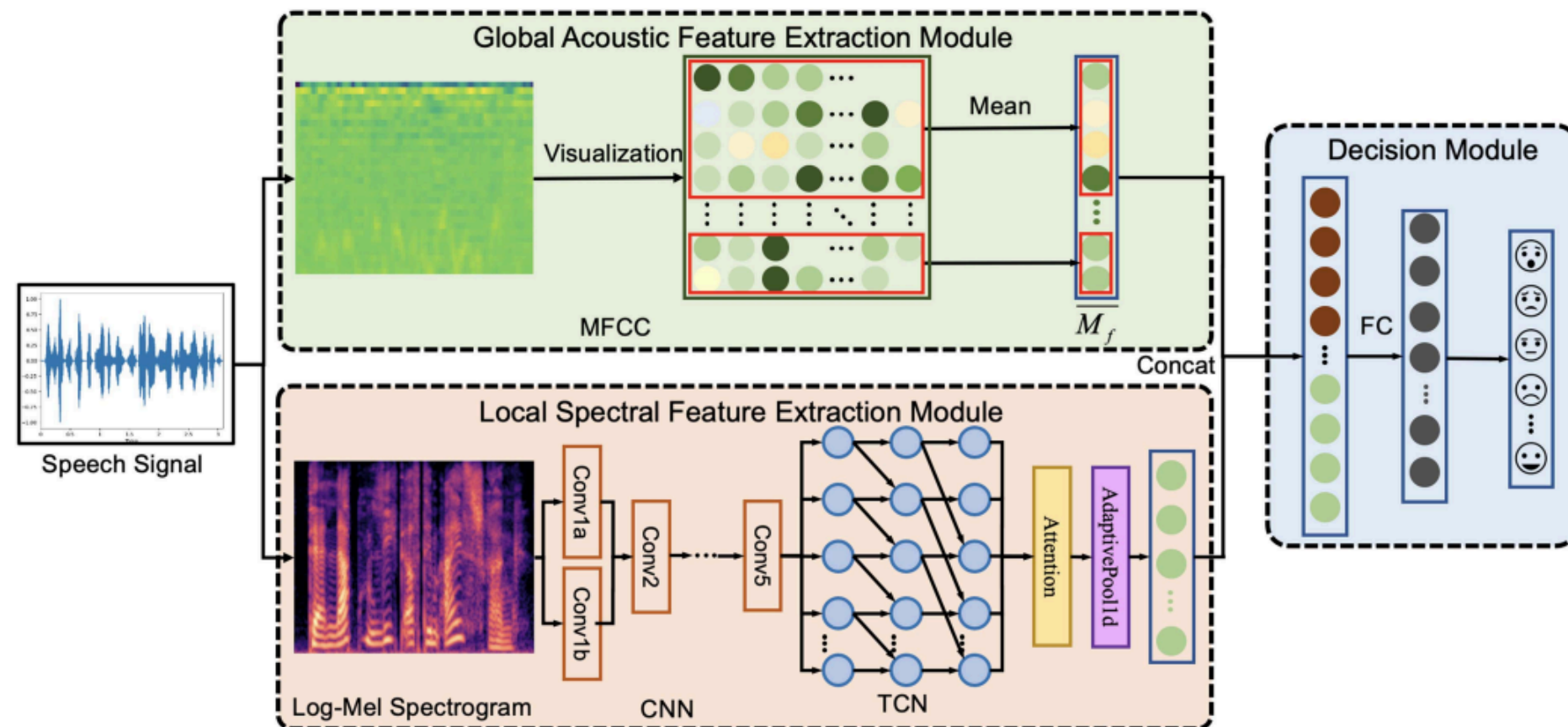
Year: 2023

Dataset:

- SpeechEmo: 475 clips from 40 Chinese drivers in simulated driving, 5 emotions, noisy environment
- EmoDB: 535 German acted utterances, 7 emotions

Methodology: Proposed GL-DSED model, combining global features (MFCC-based) with local features (CNN + Temporal Convolutional Network + self-attention). Fused features in a decision module.

Model Used: Global-Local Driver Speech Emotion Detection (GL-DSED) Network.



Literature Survey

How it Works:

This model is optimized for noisy environments like vehicles, combining both global and local features of speech.

Global Feature Branch: Extracts overall characteristics using MFCC-based features to capture the general tone of speech.

Local Feature Branch: Uses a CNN, a Temporal Convolutional Network (TCN), and a self-attention mechanism to detect short-term, local emotional variations.

Feature Fusion: Features from both branches are fused in the final decision module to achieve a more robust and accurate emotion prediction by considering both detailed and contextual information.

Performance:

EmoDB: 87.6% UA, 87.4% WA → better than BLSTM-DSA and 3D-ARCNN.

SpeechEmo: 33% UA, 29.9% WA, still outperforming baselines under noisy driving-like conditions.

Summary: By fusing global and local cues, the GL-DSED network achieved state-of-the-art SER accuracy. The introduction of the SpeechEmo dataset contributes a valuable resource for driver-focused affective computing in intelligent cockpits.

Literature Survey

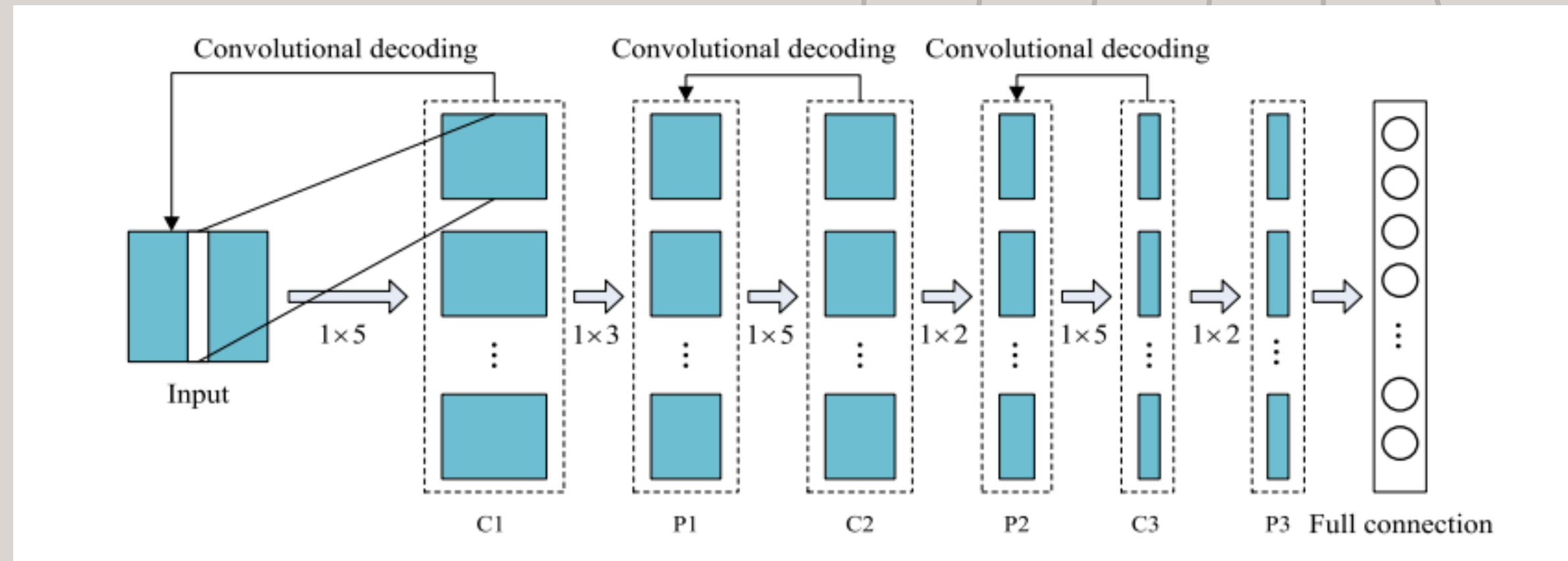
Intelligent Emotion Detection Method Based on Deep Learning in Medical and Health Data

Author(s): Xin Li, Mingwei Lin, Yongbo Xue
Year: 2021

Dataset: Medical and health-related speech/text datasets (patient interactions, multimodal records; splits for training/testing).

Methodology: Designed a deep learning model combining CNN (local feature extraction), RNN/LSTM (temporal modeling), and attention mechanisms to detect emotions in healthcare data.

Model Used: Hybrid deep learning model combining CNN, RNN/LSTM, and attention mechanism.



Literature Survey

How it Works:

This model processes multimodal medical data (e.g., text and speech) through a sequential architecture.

CNN for Local Features: Extracts key local patterns from the input data.

RNN/LSTM for Temporal Modeling: Captures sequential dependencies and contextual flow over time.

Attention Mechanism: Highlights the most relevant parts of the input sequence, enabling the model to focus on critical moments indicative of emotional states. The combination of these components allows for precise and context-aware emotion detection in healthcare settings.

Performance: Achieved >90% accuracy, outperforming traditional ML models (SVM, decision tree, k-NN) by 7-12%, and showed robustness across multimodal data.

Summary: The method enables accurate detection of patient emotions for healthcare, improving decision support, remote diagnosis, and patient-doctor communication.

DATASETS

RAVDESS:

The RAVDESS Dataset contains 1440 files: 60 trials per actor x 24 actors = 1440. The RAVDESS contains 24 professional actors (12 female, 12 male), vocalizing two lexically-matched statements in a neutral North American accent. Speech emotions includes calm, happy, sad, angry, fearful, surprise, and disgust expressions. Each expression is produced at two levels of emotional intensity (normal, strong), with an additional neutral expression.

CREMA - D:

CREMA-D is a data set of 7,442 original clips from 91 actors. These clips were from 48 male and 43 female actors between the ages of 20 and 74 coming from a variety of races and ethnicities (African America, Asian, Caucasian, Hispanic, and Unspecified). Actors spoke from a selection of 12 sentences. The sentences were presented using one of six different emotions (Anger, Disgust, Fear, Happy, Neutral, and Sad) and four different emotion levels (Low, Medium, High, and Unspecified).

Proposal

System Pipeline:

Audio Preprocessing: Adaptive signal enhancement (SNR estimation, Spectral Subtraction) and segmentation (VAD) for varied audio conditions.

Feature Extraction: Log-Mel Spectrograms, including Delta and Delta-Delta coefficients, for rich spectral and temporal features.

Deep Learning Model: A hybrid CNN-BiLSTM-Transformer with an attention network to capture short-range temporal and long-range dependencies.

Classification: Six standardized emotion categories: neutral, happy, sad, angry, fear, disgust.

Evaluation: Benchmarked on RAVDESS and CREMA-D with speaker-independent splitting, measuring accuracy and Weighted F1-score.

Objectives:

Achieve high accuracy and speaker-independent recognition across combined datasets.

Utilize and compare performance of the robust hybrid CNN-BiLSTM-Transformer model.

Use data augmentation (pitch shift, time stretch, noise injection) and regularization (Dropout, weight decay) for better generalization.

Applications:

Healthcare – mental health monitoring.

Customer Service – emotion-aware call centers.

Education – adaptive learning systems.

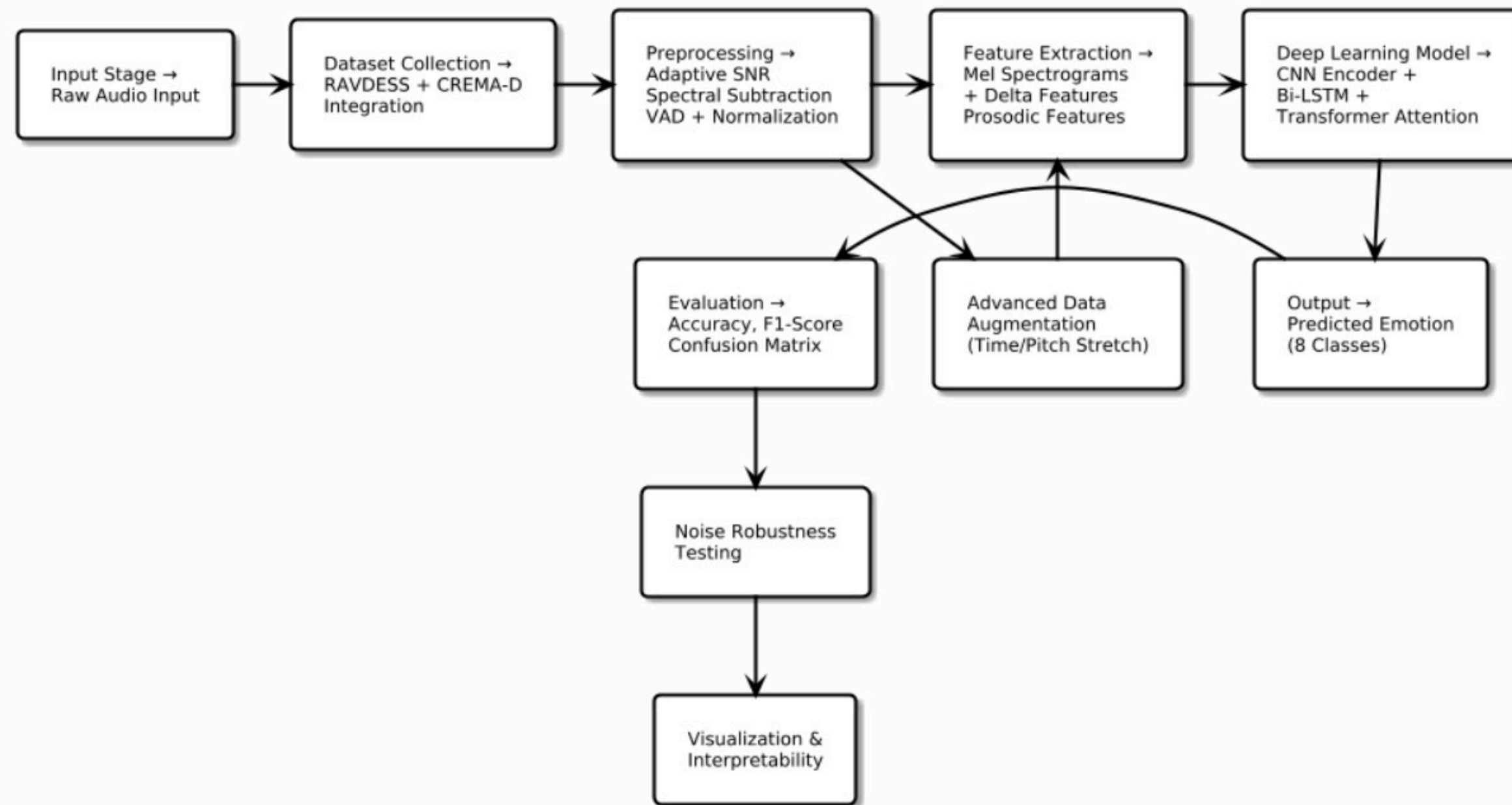
Safety – driver emotion monitoring.

Requirements:

Software: Python with PyTorch/LibROSA/Scikit-learn.

Hardware: CUDA-enabled GPU (e.g., NVIDIA RTX series) for efficient training.

Architecture Diagram



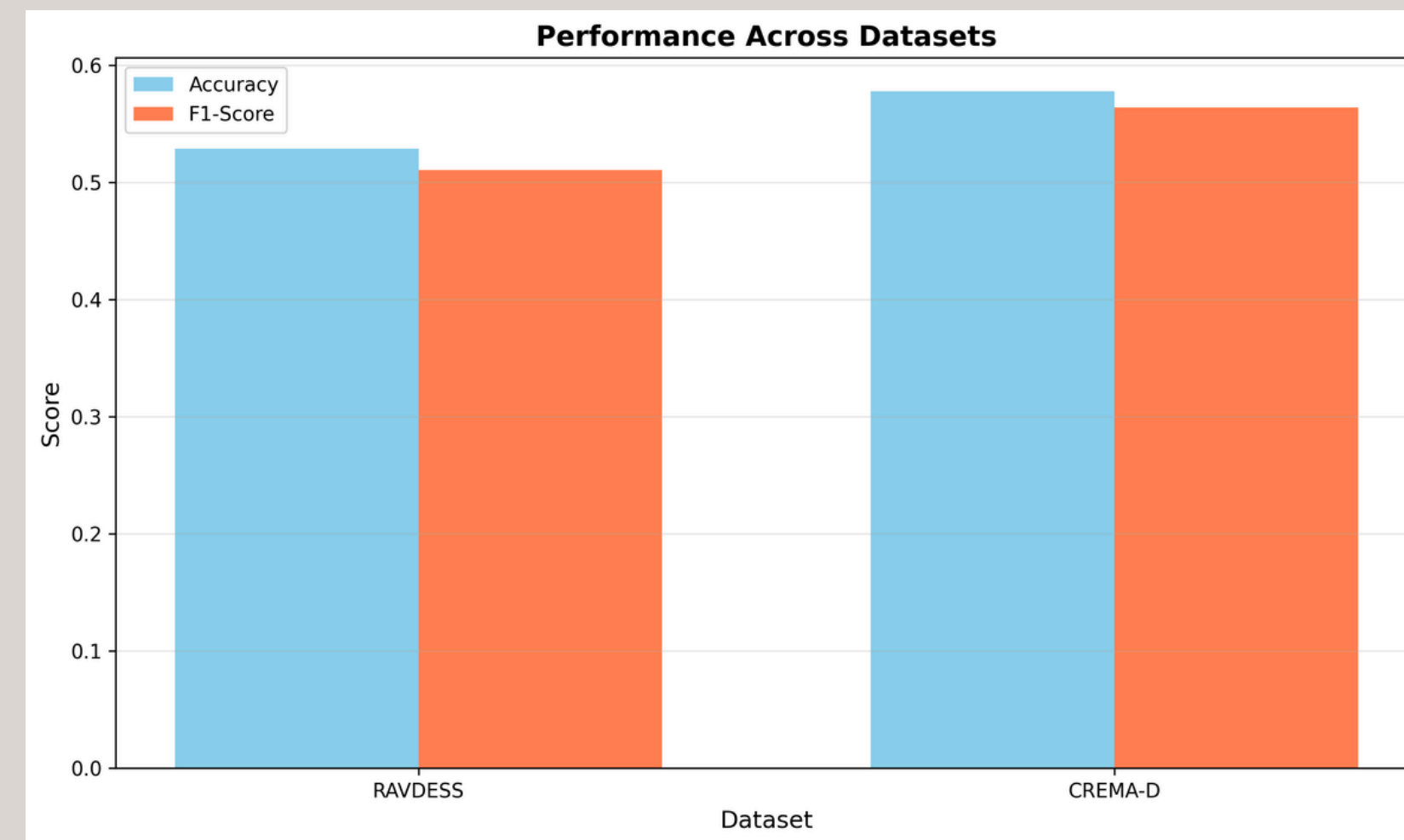
Implementation

This project implements a robust Speech Emotion Recognition (SER) system utilizing a hybrid deep learning architecture designed for cross-speaker generalization. The pipeline begins with advanced, adaptive audio preprocessing on combined RAVDESS and CREMA-D datasets, employing SNR estimation and spectral subtraction to clean low-quality audio, alongside Voice Activity Detection to segment the speech. Features are extracted as 3-channel Log-Mel Spectrograms (Mel + Delta + Delta-Delta) and augmented on-the-fly during training with techniques like pitch shift, time stretch, and noise injection. The core of the system is a CNN-BiLSTM-Transformer model, where the CNN extracts local features, the BiLSTM captures short-term temporal context, and the Transformer's attention mechanism models global, long-range dependencies for the final classification into six standardized emotions. The model is trained using PyTorch with an Early Stopping mechanism and rigorously evaluated using speaker-independent data splits to ensure high Accuracy and Weighted F1-Scores on the test set.

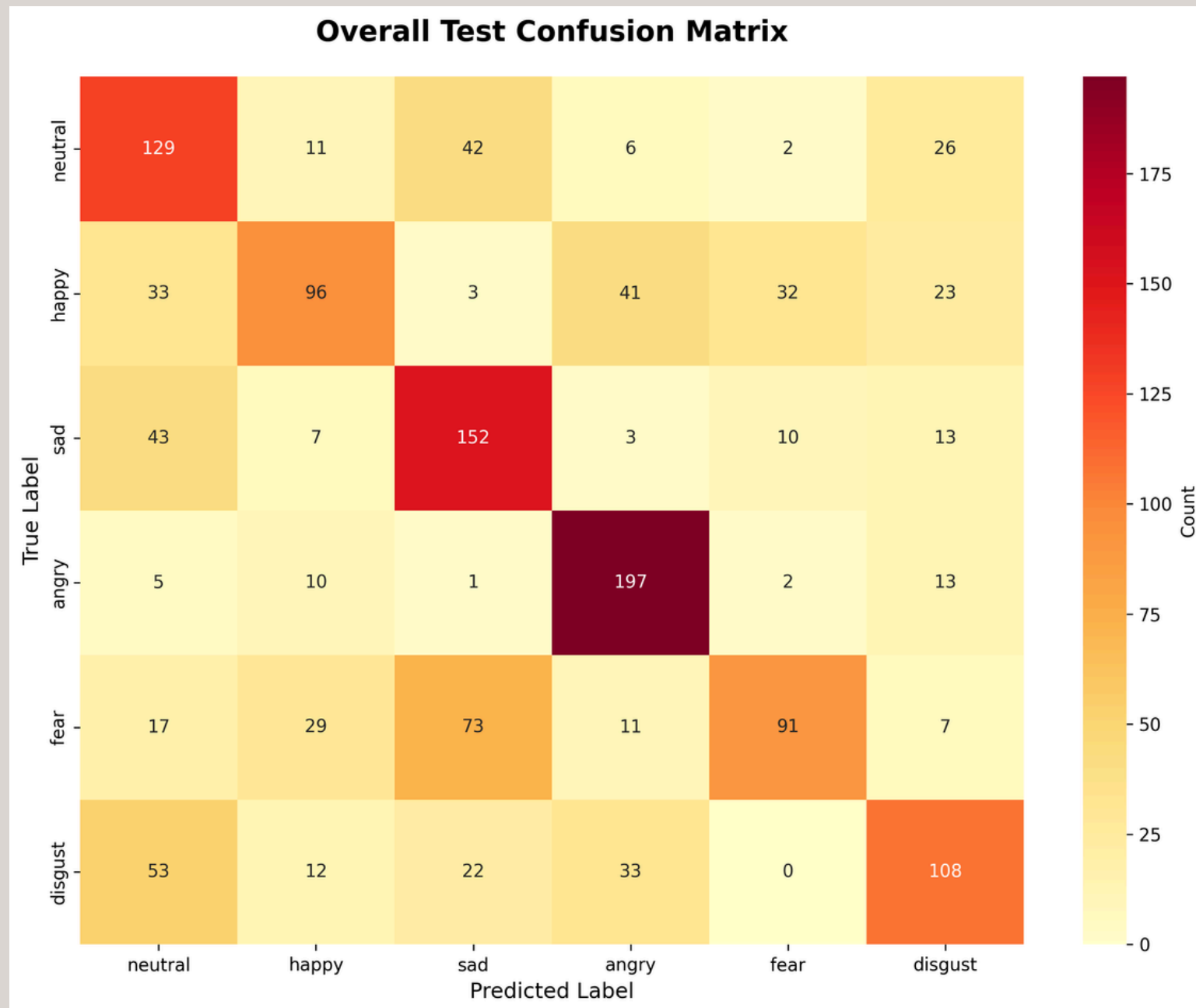
Comparison Result

The proposed model demonstrated moderate performance in the task of Speech Emotion Recognition (SER). Its best overall result was a test accuracy of 57.01% (F1 0.561) pooled across both datasets, with 57.75% on CREMA-D alone and 52.88% on RAVDESS alone. However, it showed a clear generalization gap: accuracy dropped to 30.93% when trained on CREMA-D and cross-tested on RAVDESS, and further to 17.40% in the reverse direction (trained on RAVDESS, tested on CREMA-D).

In sharp contrast, the developed model (GL-DSED), showcased a vastly superior performance on the same core task. When tested on the EmoDB dataset, the developed model achieved an Unweighted Accuracy (UA) of 87.61%. This represents an improvement of roughly 30.6 percentage points compared to the proposed model's overall test accuracy.



Comparison Result



Comparing the proposed model's SER results with the developed model's Fatigue Classification highlights a massive difference in task difficulty and model effectiveness. The proposed model's performance ranged from a failure state of 17.40% (cross-dataset) up to a moderate 57.75% accuracy.

Conversely, the developed Fatigue Classification model exhibited consistently high and robust performance, with accuracies ranging from a low of 79% (for Type 3 Fatigue) up to a high of 90% (for Type 2 Fatigue). Even the worst-performing class in the fatigue model 79% is significantly higher than the best overall accuracy achieved by the proposed model's accuracy of 57.75%.

Future Scope

- **Real-Time Implementation:** Extending the system for low-latency, real-time emotion recognition, enabling deployment in interactive applications such as virtual assistants, online learning platforms, and healthcare monitoring.
- **Multilingual Support:** Expanding the model to handle speech data from multiple languages and dialects, ensuring wider applicability across diverse cultural and linguistic settings.
- **Audio-Video Fusion:** Integrating facial expression and body-language analysis with speech signals to build a multimodal system that improves accuracy and robustness in emotion recognition.

Conclusion

Technical Aspect:

Innovative Framework: Integrated transformer-based architectures with intelligent signal processing for real-world accuracy

Advanced Architecture: Attention-driven models with automatic emotional pattern discovery and interpretable insights

Performance & Impact:

Evaluated across two demographically distinct datasets (RAVDESS, CREMA-D) with systematic augmentation, revealing solid in-domain performance alongside a measurable cross-corpus generalization gap

Healthcare & Education Applications: Enables mental health monitoring, adaptive learning, and enhanced human-AI collaboration

Future Directions:

Privacy-Preserving Technologies: Federated learning for secure emotion recognition

Cross-cultural AI Integration: Culturally aware systems with large language model integration

Edge Computing Deployment: Mobile and IoT optimization for ubiquitous emotion recognition

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The background is a light gray color. It features several abstract, thin gray lines. In the top left, there are wavy, organic lines. In the top right, there is a prominent spiral pattern that winds inward. In the bottom left, there are more wavy lines, some forming a partial circular shape. In the bottom right, there is a single, smooth wavy line.

**Thank
You!**